

TCC805x MCU BSP

Release Note

V1.0.4 [A]

2021-06-10

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1 INTRODUCTION

1.1 Feature

- Support Cortex-R5 MCU BSP
- Support two types of RTOS (FreeRTOS and uC/OS-III) via System Adaptation Layer (SAL) single interface
- Support Interrupt Driver
- Support Timer Driver
- Support Clock Driver
- Support Controller Area Network (CAN) Controller Driver
- Support Local Interconnect Network (LIN) Driver
- Support Mailbox Driver
- Support Memory Access Identifier Filter Driver
- Support Safe Display Manager Driver
- Support Default Slave Error Interrupt Handler Driver
- Support Fault Management Unit Driver
- Support General Direct Memory Access Controller Driver
- Support UART Driver
- Support I2C Driver
- Support SPI Driver
- Support ADC Driver
- Support Port (GPIO) Driver
- Support Input Capture Time Counter (ICTC) Driver
- Support PWM Driver
- Support Windowed Watchdog Timer (WDT) Driver
- Support Inter-Process Communication (IPC) Driver
- Support Power Manager IO (PMIO) Driver

1.2 System Requirements

1.2.1 Hardware Requirements

- TCC805x Evaluation Board (EVB)

Board	Board Name/Version
CPU Board	TCC8050_53_LPD4X322_SV0.1
	TCC8059_LPD4X321_SV0.1
Main Board	TCC805X_6X_MAIN_CV0.11

- 12V/10A Power Adapter
- USB Cable

1.2.2 Software Requirements

- VTC Driver
- FWDN

1.3 Supported Operating System

This chapter describes the supported OS.

Operating Environment	Description
OS	FreeRTOS (V10.3.1), uC/OS-III (V3.04.04)
Device	TCC805x (TCC8059, TCC8050, TCC8053) EVB
Firmware	TCC805x MCU BSP

- The SDK supports uC/OS-III version 3.04.04, but the distributed SDK does not include the content of uC/OS-III itself. Visit www.micrium.com and you must comply with the license policy to use uC/OS-III RTOS for commercial product.

1.4 Release Components

This chapter describes the released components in this version.

Component	Description
Source Code	MCU BSP release git (ssh://git.telechips.com/mcu/mcu_bsp.git) ■ git branch -a ■ git checkout tcc805x_release_vx.x.x (x means the released version, such as 1.0.0)
RTOS	FreeRTOS (V10.3.1)

Note: If you have any problem about accessing the git repository. Contact Telechips: sales@telechips.com.

1.5 Document List

Folder	Document	Remark
TCC805x MCU BSP-Getting Started		
TCC805x MCU BSP-Release Note		
SW	TCC805x Automotive Security Solution-User Guide for HSM	
	TCC805x MCU BSP-API Specification for Analog Digital Converter Controller	
	TCC805x MCU BSP-API Specification for Clock Control Unit	
	TCC805x MCU BSP-API Specification for Controller Area Network Controller	
	TCC805x MCU BSP-API Specification for Default Slave Error Interrupt Handler	
	TCC805x MCU BSP-API Specification for Fault Management Unit	
	TCC805x MCU BSP-API Specification for Firmware Upgrade	
	TCC805x MCU BSP-API Specification for General Direct Memory Access Controller	
	TCC805x MCU BSP-API Specification for General Purpose Input Output	
	TCC805x MCU BSP-API Specification for General Purpose Serial Bus Controller	
	TCC805x MCU BSP-API Specification for Input Capture Time Counter	
	TCC805x MCU BSP-API Specification for Inter-Integrated Circuit Master Controller	
	TCC805x MCU BSP-API Specification for Inter Processor Communication	
	TCC805x MCU BSP-API Specification for Interrupt Controller	
	TCC805x MCU BSP-API Specification for Local Interconnect Network	
	TCC805x MCU BSP-API Specification for Mailbox	
	TCC805x MCU BSP-API Specification for Memory Access Identifier Filter	
	TCC805x MCU BSP-API Specification for Power Management Input Output	
	TCC805x MCU BSP-API Specification for Pulse Density Modulation	
	TCC805x MCU BSP-API Specification for Safe Display Manager	
	TCC805x MCU BSP-API Specification for System Adaptation Layer	
	TCC805x MCU BSP-API Specification for Timer	
	TCC805x MCU BSP-API Specification for Universal Asynchronous Receiver Transmitter	
	TCC805x MCU BSP-API Specification for Windowed Watchdog Timer	
	TCC805x MCU BSP-User Guide	

2 RELEASE INFORMATION

2.1 MCU BSP V1.0.4 (2021-06-10)

2.1.1 New Feature/Improvement

- Improved SAL interface
 - The description of parameters on SAL_EventGet is updated, especially puiFlags.
 - The remark of SAL_TaskSleep is added related to 1 ms on uiMillSec. Refer to "*TCC805x MCU BSP-API Specification for System Adaptation Layer*".
- Improved ICTC driver
 - Removed static keyword for some ICTC device driver APIs for use in application.
- Improved PDM (PWM) driver
 - PDM peripheral clock is changed to 25 MHz from 400 MHz.
- Improved PMIO driver
 - Improved power communication for STR mode. Refer to "*TCC805x MCU BSP-API Specification for Power Management Input Output*".
- Improved Mailbox driver
 - Added reserved device for communication with U-Boot layer of AP core. Refer to "*TCC805x MCU BSP API Specification for Mailbox*".
- Improved IPC buffer control
 - Fixed the calculation error about ring buffer index when the buffer is full.
- Improved boot time by modifying the sequence of copying RAM data (.data, .bss) from S-NOR to SRAM memory after enabling I/D cache.
- Updated the image making tools for R5 MCU's firmware to the latest version.
 - Updated "tcmktool" for Linux.
 - Updated "tcmktool.exe" for Windows.
- Improved a mechanism for setting of RTOS timer tick.
 - The mechanism for setting the timer is improved by referring to the RTOS's configuration value such as "configTICK_RATE_HZ" and "OS_CFG_TICK_RATE_HZ". (Default value is 1000)
 - Tick rate in Hz is valid between 10 (100 ms) and 1000 (1 ms).

2.2 MCU BSP V1.0.3 (2021-04-09)

2.2.1 New Feature/Improvement

- SAL_GetTickCount() function is modified in FreeRTOS environment so that it can be called in ISR mode.
- Power Management Input Output (PMIO) driver is upgraded for Suspend To RAM (STR) mode. Refer to "TCC805x MCU BSP-API Specification for Power Management Input Output" for details.
- System memory configuration is modified for non-cacheable area via MPU driver.

2.3 MCU BSP V1.0.2 (2021-01-22)

2.3.1 New Feature/Improvement

- Support Cortex-R5 processor in TCC805x CS version only
- Upgrade the following device drivers
 - UART Driver
 - PORT (GPIO) Driver
 - Power Management Input Output (PMIO) Driver
- Added system reset control
 - Added pmu_reset module

2.4 MCU BSP V1.0.0 (2020-11-27)

2.4.1 New Feature/Improvement

- Official release of TCC805x MCU BSP
- Default RTOS of SDK is set to FreeRTOS
- Available only for TCC805x ES version
- Integrated with System Adaptation Layer (SAL)
- Support the following device drivers
 - Interrupt Driver
 - Timer Driver
 - Clock Driver
 - UART Driver
 - I2C Driver
 - SPI Driver
 - ADC Driver
 - PORT (GPIO) Driver
 - Input Capture (ICTC) Driver
 - PWM Driver Supported
 - Windowed Watchdog Timer (WDT) Driver
 - CAN Driver
 - Default Slave Error Interrupt (DSEI) Driver
 - Fault Management Unit (FMU) Driver
 - General Direct Memory Access (GDMA) Driver
 - Mailbox Driver
 - LIN Driver
 - Memory Access Identifier Filter (MFID) Driver
 - Power Management Input Output (PMIO) Driver
 - Hardware Security Manager (HSM) Driver
- Support the following system managers
 - Safe Display Manager (SDM)
 - Hardware Security Manager (HSM)
 - Inter Processor Communication (IPC)

2.4.2 Notification

- SDK supports uC/OS-III version V3.04.04, but the distributed SDK does not include the content of uC/OS-III itself. Visit www.micrium.com and must comply with the license policy to use uC/OS-III RTOS for commercial product.

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